

Notes: Dávila (RES, 2020)

Using Elasticities to Derive Optimal Bankruptcy Exemptions

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1 Big picture

- **Question.** How should bankruptcy exemptions be set when households borrow using unsecured credit, can default, and value insurance against bad states?
- **Core object.** The paper derives the marginal welfare effect of raising the exemption level m , measured in date-0 dollars and normalized by marginal utility at date 0.
- **Trade-off.** A higher exemption improves insurance for bankrupt households but worsens credit supply by reducing lenders' expected recovery.
- **Empirical application.** The paper applies the formula to U.S. state exemption laws using bankruptcy filings, RateWatch, PSID, ACS, BLS, and FHFA data.
- **Main result.** Increasing exemptions is estimated to raise welfare on average, with large heterogeneity across states and income quintiles.

2 Incremental contribution

- Relative to quantitative bankruptcy models, the paper derives policy formulas that apply to a broad class of models and require only a small set of observables.
- Relative to empirical bankruptcy studies, it translates behavioral and price moments into an explicit normative welfare test for exemption levels.
- Relative to sufficient-statistics public finance, it extends the Chetty/Saez logic to defaultable debt, bankruptcy exemptions, and household credit markets.
- It identifies the marginal credit-supply response, the probability of asset bankruptcy, and state-contingent marginal valuation as the key empirical objects.

3 Data and measurement

Bankruptcy exemptions determine how much property a debtor may keep after filing. Homestead exemptions protect home equity, while personal exemptions protect other assets. The empirical implementation combines ID-FJC bankruptcy filings, RateWatch interest rates, PSID household data, state exemption data, ACS income data, BLS unemployment rates, and FHFA house prices.

Interpretation. The empirical variation comes from state-level exemption rules, income-quintile heterogeneity, and local credit-market conditions.

4 Models and empirical specifications

4.1 Baseline sufficient-statistics formula

$$\frac{dW/dm}{u'(c_0)} = \frac{\partial q_0}{\partial m} b_1 + \pi_m E_m \left[\frac{\beta u'(c_1^D)}{u'(c_0)} \right].$$

Interpretation. The first term is the marginal cost: a higher exemption worsens the loan price q_0 , and this cost scales with debt b_1 . The second term is the marginal insurance benefit: bankrupt households who claim the full exemption receive an extra dollar, valued using their stochastic discount factor.

4.2 Sufficient-statistics corollary

$$\text{Required objects: } b_1, \quad \frac{\partial q_0}{\partial m}, \quad \pi_m, \quad E_m \left[\frac{\beta u'(c_1^D)}{u'(c_0)} \right].$$

Interpretation. The formula avoids solving the full optimal default and borrowing problem. Optimal borrowing and default choices enter only indirectly through the sufficient statistics.

4.3 Credit-supply regression

$$\log(\text{gross interest rate})_{bct} = \alpha + \beta \text{Exemption}_{st} + \text{Controls}_{ct} + \text{YearFE} + \text{LocationFE} + \varepsilon_{bct}.$$

Interpretation. The coefficient on exemptions identifies how much loan rates rise when exemptions increase. This maps into the model's credit-supply derivative.

5 Identification and empirical design

The paper is primarily a measurement and sufficient-statistics exercise. Its credibility comes from deriving a welfare formula in a broad class of models and then measuring the required objects. The key assumptions are that households optimize, lenders are competitive or profits are invariant, and observed interest-rate responses capture the relevant credit-supply sensitivity.

6 Tables and figures

6.1 Figure 1: Borrowers' date 1 consumption and bankruptcy decision; Figure 2: Lenders' payment

Read: Figure 1 shows repayment, forced default, and strategic default. Figure 2 shows creditor recovery across the same endowment states.

Economic message. These figures translate exemption policy into borrower and lender payoffs.

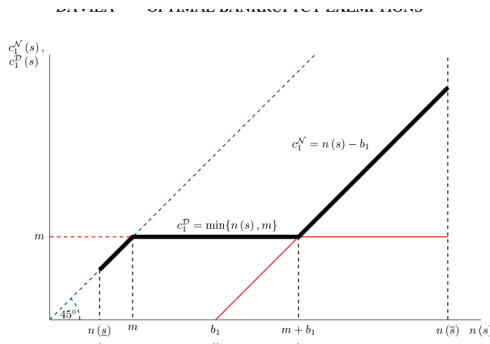


FIGURE 1
Borrowers' date 1 consumption and bankruptcy decision

Notes: This figure illustrates the optimal default decision made by households for a given realization of households' date 1 endowment $n(s)$.

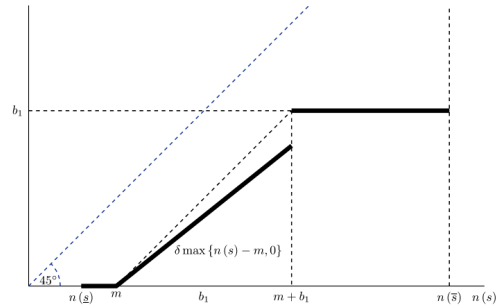


FIGURE 2
Lenders' payment

Notes: This figure illustrates the repayment received by lenders for a given realization of households' date 1 endowment $n(s)$. The discontinuity in this figure is due to the presence of deadweight losses associated with default.

(a) Borrowers' date 1 consumption and bankruptcy decision

(b) Lenders' payment

6.2 Figure 3: Comparative statics $m^*(\delta)$ and $m^*(\sigma)$

Read: The optimal exemption rises with lender recovery δ and household risk σ .

Economic message. Higher recovery lowers credit-supply costs; higher uncertainty raises insurance benefits.

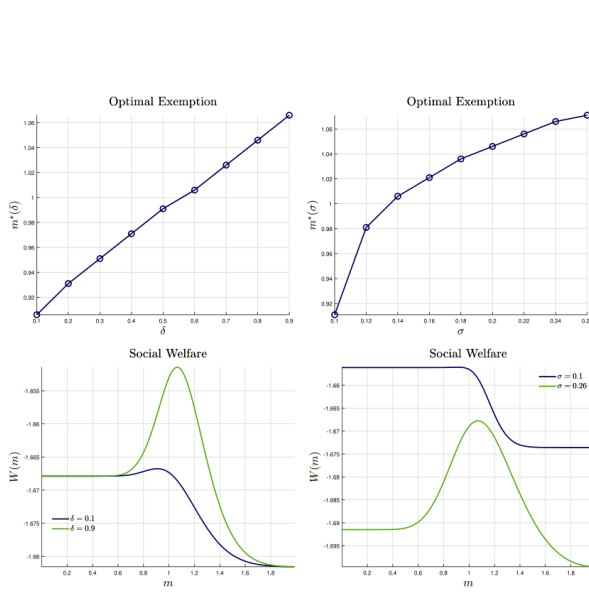


FIGURE 3
Comparative statics $m^*(\delta)$ and $m^*(\sigma)$

Notes: The left panel shows how the optimal exemptions m^* , defined in equation (2.14), varies as a function of the recovery rate by lend in case of default δ (the complement of the dead-weight loss). The single period is meant to represent 3 years, with parameters $\beta = 0.9$, $r^L = 1.04^3 - 1$, $\gamma = 2$, $\eta_0 = 1$, $\mu = 0.15$, $\sigma = 0.18$, $\xi = 0.2$, and $\bar{s} = 2$. I consider a range for the recovery rate δ given by $\delta \in [0.1, 0.9]$. The right panel shows how the optimal exemptions m^* , defined in equation (2.14), varies as a function of the volatility of the distribution

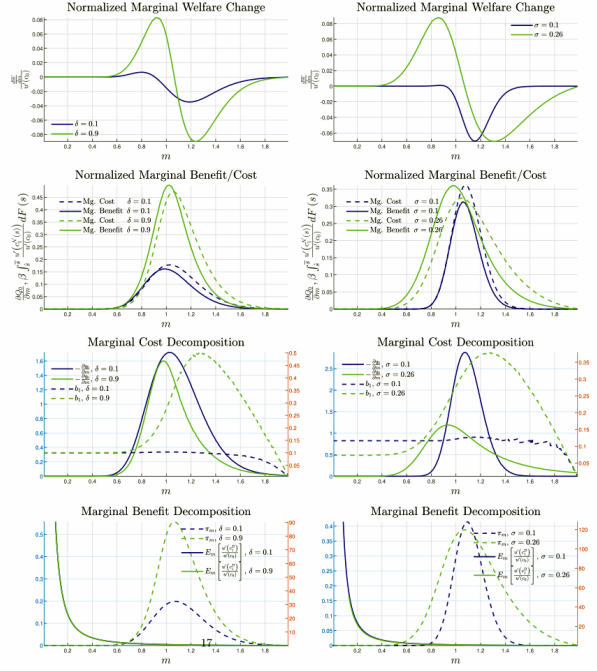


FIGURE 4
Marginal Welfare Change and Marginal Cost/Benefit Decomposition

Notes: Each row of plots in Figure 4 respectively shows, from top to bottom: i) the normalized social welfare change as a function of m , defined in equation (2.12), ii) the normalized marginal benefit, $\frac{\partial W}{\partial m}$, and cost, $\beta \int_0^1 \frac{\partial u(c^T)}{\partial m} dF(s)$, iii) the components of the normalized marginal benefit, $-\frac{\partial \eta_0}{\partial m}$ and b_1 , and iv) the components of the normalized marginal cost, π_m and $E_m \left[\frac{\partial u(c^T)}{\partial m} \right]$. All left figures have parameters $\beta = 0.92^3$, $r^L = 1.04^3 - 1$, $\gamma = 2$, $\delta = [0.1, 0.9]$, $\eta_0 = 1$, $\mu = 0.2$, $\sigma = 0.18$, $\xi = 0.2$, and $\bar{s} = 2$. All right figures have parameters $\beta = 0.92^3$, $r^L = 1.04^3 - 1$, $\gamma = 2$, $\delta = 0.75$, $\eta_0 = 1$, $\mu = 0.2$, $\sigma = [0.1, 0.26]$, $\xi = 0.2$, and $\bar{s} = 2$.

6.3 Figure 4: Marginal welfare change and marginal cost/benefit decomposition

Read: The figure decomposes dW/dm into marginal costs, marginal benefits, and their component sufficient statistics.

Economic message. This is the visual version of the paper's main formula.

6.4 Table 1: Empirical counterparts; Table 2: Credit supply sensitivity

Read: Table 1 maps theory to data. Table 2 estimates the interest-rate effect of exemptions.

Economic message. Credit-supply sensitivity is the main empirical cost term.

TABLE 1
Empirical counterparts

Measured variable	Data source	Maps into
Interest-rate sensitivity	RateWatch, Exemptions Dataset, Regional Controls (ACS/BLS/FHFA)	Q_m
Households' liabilities	PSID	Q_m
Households' expenditure	PSID	G
Income transition matrix	PSID	Π
Prob. of bankruptcy	ID-FIC	$\Pi_{N' \rightarrow m}, \Pi_{N'}$
Prob. of no-asset bankruptcy	ID-FIC	$\Pi_{N' \rightarrow m}$

Notes: This table schematically describes the data sources used to calculate each sufficient statistic required to compute $\frac{d\hat{W}}{dm}$, as described in Proposition 3.

TABLE 2
Credit supply sensitivity

	Dependent variable: log(gross interest rate)						
	Personal (1)	Personal (2)	Personal (3)	Auto (4)	Auto (5)	Auto (6)	Mortg: (7)
Exemption (homestead)	0.0071* (0.0041)		0.0044 (0.0079)	0.0058** (0.0025)		0.0054 (0.0037)	-0.0001 (0.0005)
Exemption (personal)		0.0185 (0.0538)			0.0240 (0.0201)		
Unemployment rate	0.0003 (0.0004)	0.0003 (0.0004)	0.0005 (0.0004)	0.0003* (0.0002)	0.0003 (0.0002)	0.0004* (0.0002)	0.0000 (0.0001)
House prices	-0.00002** (0.00001)	-0.00002** (0.00001)	-0.00002*** (0.00001)	0.000005 (0.000003)	0.000004 (0.000003)	0.000002 (0.000003)	0.0000 (0.0000)
Income	0.0023 (0.0017)	0.0024 (0.0017)	0.000002 (0.0018)	0.00004 (0.0009)	0.0001 (0.0009)	0.0008 (0.0011)	-0.0007 (0.0005)
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Branch FE	Yes	Yes	No	Yes	Yes	Yes	Yes
County FE	No	No	Yes	No	No	No	No
Observations	16,255	16,255	16,255	18,349	18,349	18,349	7,73
Adjusted R ²	0.8348	0.8348	0.3614	0.8885	0.8884	0.4473	0.8452

Notes: This table reports the results of running Regression R1 using personal unsecured loans' interest rates, auto loan interest rates, and mortgage rates (30-year fixed rate mortgage) as the dependent variable. Exemptions are measured in millions of dollars. Interest rates are yearly averages from RateWatch, county unemployment rate is from the BLS, county house prices are from the FHFA, and county median household income is from the ACS. Standard errors, in parentheses, are clustered at the state level, which allows for serial correlation in the error term. ***Significant at the 1% level. **Significant at the 5% level. *Significant at the 10% level.

(a) Empirical counterparts

(b) Credit supply sensitivity

6.5 Figure 5: Marginal welfare changes and exemption levels; Table 3: Marginal welfare changes

Read: Figure 5 plots average state-level gains. Table 3 summarizes state-quintile distributions of dynamic and flow marginal welfare gains.

Economic message. Welfare gains are positive on average but highly heterogeneous.

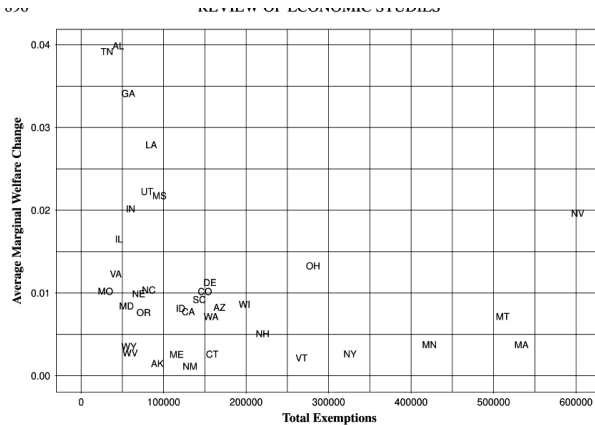


FIGURE 5
Marginal welfare changes and exemption levels

Notes: This figure shows average normalized marginal welfare changes associated with a one-dollar increase in the exemption level l

TABLE 3
Marginal welfare changes

	$\frac{dW}{d\alpha}$ (marginal welfare change)					F (flow change)				
Quintile	Mean	S.D.	Pctl (10)	Median	Pctl (90)	Mean	S.D.	Pctl (10)	Median	Pctl (9)
First	0.0038	0.0034	0.0009	0.0026	0.0079	0.016	0.0015	0.0003	0.0011	0.003
Second	0.0050	0.0040	0.0010	0.0038	0.0117	0.0023	0.0022	0.0004	0.0017	0.005
Third	0.0039	0.0028	0.0013	0.0028	0.0062	0.0020	0.0018	0.0003	0.0014	0.003
Fourth	0.0046	0.0038	0.0015	0.0034	0.0100	0.0019	0.0015	0.0006	0.0015	0.004
Fifth	0.0063	0.0045	0.0021	0.0051	0.0129	0.0016	0.0011	0.0004	0.0013	0.003
Overall	0.0048	0.0041	0.0012	0.0035	0.0112	0.0019	0.0017	0.0004	0.0013	0.004

Notes: This table reports summary statistics of the distribution of normalized marginal welfare changes by state associated with a one-dollar increase in the exemption level. Average normalized welfare changes are calculated as an unweighted sum of normalized marginal welfare changes across different income levels, calculated following Proposition 3. The determination of income quintiles by state uses income data from the 2013 5-year ACS. The figure includes 36 U.S. states. It excludes states with unlimited exemption levels, states with dominated state exemptions, as well as two states for which some income quintile is not populated in the PSID data (Rhode Island and North Dakota). Preference parameter $\beta = (0.94)^2$ and $\gamma = 2$.

(a) Marginal welfare changes and exemption levels

(b) Marginal welfare changes

6.6 Figure 6: Evolution of personal bankruptcy filings (2001–17)

Read: The figure shows the number of filings and the Chapter 7/Chapter 13 decomposition.

Economic message. Bankruptcy policy affects a large and time-varying set of households.

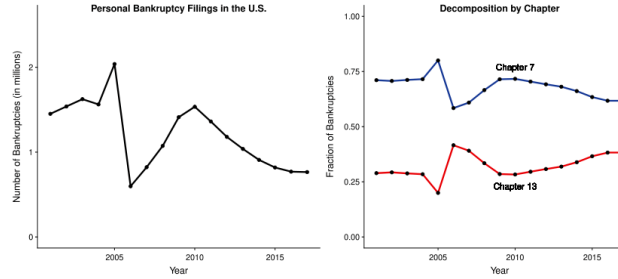


FIGURE 6

Evolution of personal bankruptcy filings (2001–17)

Notes: The left panel shows the yearly number of personal bankruptcy filings in the U.S. from 2001 until 2017, as reported by the U.S. Department of Justice through the ID-FJC. The right panel shows the fraction of Chapter 7 and Chapter 13 filings over the same period. See Livshits *et al.* (2010, 2016) for a systematic analysis of the long-run evolution of consumer bankruptcies, starting in 1970.

Evolution of personal bankruptcy filings (2001–17)

6.7 Tables 4–8: Data construction summary statistics

Read: Tables 4–8 describe bankruptcy filings, debtor balance sheets, exemption levels, interest rates, regional controls, and PSID household information.

Economic message. These tables show how the sufficient statistics are measured.

TABLE 4
Bankruptcy filings by chapter and no-asset status

	Chapter 7 (%)	Chapter 13 (%)	Both chapters (%)
No-asset cases	67.55	0.11	67.66
Asset cases	3.12	29.21	32.33
Both cases	70.67	29.32	

Notes: This table reports the share of bankruptcies by chapter and by “no-asset” denomination in the U.S. from 2001 until 2016, as reported by the U.S. Department of Justice through the ID-FJC. A bankruptcy filing is defined as an asset case when “there will be funds available for distribution to unsecured creditors, after any exempt property is excluded

(a) Bankruptcy filings by chapter and no-asset status

TABLE 6
Summary statistics of bankruptcy exemption levels

	Mean	St. Dev.	Pctl (10)	Median	Pctl (90)
Homestead (2008)	113,493	139,083	10,000	63,750	300,000
Personal (2008)	17,641	10,650	6,280	15,150	28,945
Total (2008)	131,134	141,787	29,290	88,150	324,080
Homestead (2016)	147,499	151,811	27,893	97,375	423,000
Personal (2016)	24,109	13,292	11,700	20,300	35,960
Total (2016)	171,608	156,038	45,560	122,250	448,840

Notes: This table reports summary statistics for the level of personal, homestead, and total state-level exemptions (defined as the sum of personal and homestead exemptions) for the years 2008 and 2016. It provides information on the mean, standard deviation, as well as the 10th and 90th percentiles of the distribution of exemptions across states. Exemptions are expressed in dollars as of the relevant year, 2008 or 2016. The summary statistics include 38 states, exclude the seven states (in addition to DC) for which homestead exemptions are unlimited and the five states in which households can opt for federal exemptions that are uniformly higher than state exemption levels.

(a) Bankruptcy exemption levels

TABLE 5
Summary statistics of bankruptcy filing information

	Mean	St. Dev.	Pctl (10)	Median	Pctl (90)	N
Net worth	−73,977	478,464	−4187,425	−37,537	12,764	9,383,7 ^a
Assets	127,948	287,923	3,720	65,442	314,232	9,383,7 ^a
Liabilities	201,925	481,458	24,199	119,837	441,378	9,383,7 ^a
Income	41,940	334,532	4,966	35,578	79,029	9,383,7 ^a

Notes: This table reports summary statistics on net worth, total assets, total liabilities, and yearly income of bankruptcy filers in the U.S. from 2008 until 2016, as reported by the U.S. Department of Justice through the ID-FJC. It provides information on the mean, median, standard deviation, as well as the 10th and 90th percentiles of each of the variables. All variables are measured in dollars and truncated to be smaller than \$100 million.

(b) Summary statistics of bankruptcy filing information

TABLE 7
Summary statistics of interest rates and regional controls

	Mean	St. Dev.	Pctl (10)	Median	Pctl (90)	N
Personal unsecured rate	0.11	0.03	0.07	0.10	0.16	313,7 ^a
Auto loan rate	0.04	0.02	0.02	0.04	0.06	207,9 ^a
Mortgage fixed rate	0.04	0.01	0.04	0.04	0.05	26.2
Unemployment rate	6.85	2.53	4.20	6.50	10.00	32.9
Median income	51,348.86	13,064.15	39,086.50	48,927.00	66,667.00	32.9
House price index	387.86	213.78	182.62	352.81	645.85	32.9

Notes: This table reports summary statistics (mean, median, standard deviation, as well as the 10th and 90th percentile of each of the variables) for the levels of net interest rates offered across branches in the U.S. between 2008 and 2016 on personal unsecured loans, auto loans on 3-year-old cars, and 30-year fixed rate mortgages. It also reports summary

(b) Interest rates and regional controls

TABLE 8
Summary statistics of households' personal information

	Mean	St. Dev.	Pctl (10)	Median	Pctl (90)	N
Income	67,512	98,266	10,957	48,050	136,500	43,6'
Expenditure	26,236	19,776	7,500	22,360	47,993	43,6'
Personal debt	6,961	41,226	0	0	17,000	43,6'
Auto loans	4,522	10,087	0	0	18,700	43,6'
Mortgage principal	50,272	98,429	0	0	175,000	43,6'

Notes: This table presents summary statistics for the sample of 43,673 households obtained from the PSID over the period 2007–15. All variables are measured in dollars for the current year. I restrict the sample to observations with annual expenditure greater than \$100 and negative income values smaller in magnitude than \$100,000. I weight observations using the PSID core/immigrant family longitudinal weights.

Households' personal information

6.8 Tables 9–10: Default probabilities and average valuation matrix

Read: Table 9 compares Alabama and Massachusetts. Table 10 reports the average valuation matrix across income quintiles.

Economic message. Default probabilities identify who receives the insurance; the valuation matrix measures how much a marginal dollar is worth in those states.

TABLE 9
Default probabilities

Quintile	π^D		π^{mD}	
	Alabama	Massachusetts	Alabama	Massachusetts
First	0.0157	0.0068	0.6115	0.0934
Second	0.0228	0.0081	0.6359	0.1413
Third	0.0199	0.0060	0.5996	0.2433
Fourth	0.0115	0.0039	0.5933	0.3898
Fifth	0.0041	0.0007	0.6527	0.6817

Notes: This table reports two examples of yearly bankruptcy probabilities by income quintiles for the states of Alabama and Massachusetts, chosen as examples of states associated with high and low marginal welfare gains from increased exemption levels. These values are constructed combining data on the universe of bankruptcies from the Department of Justice through the ID-FJC with total population data by state from the 2013 5-year ACS. The determination of income quintiles by state uses income data also from the 2013 5-year ACS.

(a) Default probabilities

Average valuation matrix

	First	Second	Third	Fourth	Fifth
First	1	0.47	0.27	0.17	0
Second	2.64	1	0.59	0.38	0
Third	4.56	1.77	1	0.63	0
Fourth	7.35	2.93	1.65	1	0
Fifth	14.31	5.91	3.35	2.04	1

Notes: This table reports the average across the final set 36 states of their valuation matrix G . The determination of income quintiles by state uses income data from the 2013 5-year ACS. Households' risk aversion is $\gamma = 2$.

(b) Average valuation matrix

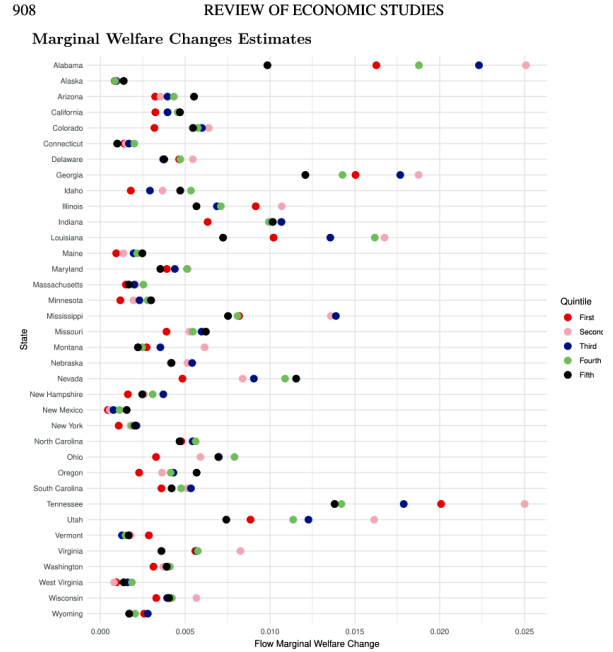
6.9 Figure 7: Marginal welfare changes by income quintile; Figure 8: Flow marginal welfare changes by income quintile

Read: Figure 7 plots normalized dynamic marginal welfare changes; Figure 8 plots flow welfare changes.

Economic message. Dynamic valuation amplifies the welfare gains relative to flow calculations.



(a) Marginal welfare changes by income quintile



(b) Flow marginal welfare changes by income quintile

7 Short summary

- The paper studies optimal bankruptcy exemptions using a sufficient-statistics approach.
- The main formula balances credit-supply costs against insurance benefits for bankrupt households.
- The required empirical objects are debt, credit-supply sensitivity, asset-bankruptcy probability, and marginal valuation in bankruptcy states.
- Increasing exemptions is estimated to be welfare-improving on average, with large heterogeneity across states and income quintiles.
- A 50,000 increase in exemptions is valued at roughly 30 billion in aggregate under the baseline calculation.

8 Conclusion

8.1 Core conclusion

Dávila shows that bankruptcy exemption policy can be evaluated using a compact set of sufficient statistics rather than a fully calibrated structural model.

8.2 Economic intuition

Higher exemptions provide insurance in bad states but raise borrowing costs by lowering creditor recovery.

8.3 Incremental contribution

The paper connects bankruptcy design to sufficient-statistics public finance and supplies the first empirical implementation of the resulting exemption formula across U.S. states and income quintiles.

8.4 Policy implication

Existing exemptions appear too low on average, but optimal policy is heterogeneous across states and income groups.

8.5 Limitations

The estimates rely on measured credit-supply elasticities, valuation matrices, and bankruptcy probabilities, and may abstract from some institutional and general-equilibrium channels.

8.6 Takeaway

Bankruptcy exemptions are an insurance instrument with measurable marginal benefits and costs.